

from various senders are the same; in reality this can be difficult to achieve. There is a considerable body of literature addressing these and other issues related to CDMA; see [Pickholtz 1982; Viterbi 1995] for details.

### 6.3    WiFi: 802.11 Wireless LANs

Pervasive in the workplace, the home, educational institutions, cafés, airports, and street corners, wireless LANs are now one of the most important access network technologies in the Internet today. Although many technologies and standards for wireless LANs were developed in the 1990s, one particular class of standards has clearly emerged as the winner: the **IEEE 802.11 wireless LAN**, also known as **WiFi**. In this section, we’ll take a close look at 802.11 wireless LANs, examining its frame structure, its medium access protocol, and its internetworking of 802.11 LANs with wired Ethernet LANs.

There are several 802.11 standards for wireless LAN technology, including 802.11b, 802.11a, and 802.11g. Table 6.1 summarizes the main characteristics of these standards. 802.11g is by far the most popular technology. A number of dual-mode (802.11a/g) and tri-mode (802.11a/b/g) devices are also available.

The three 802.11 standards share many characteristics. They all use the same medium access protocol, CSMA/CA, which we’ll discuss shortly. All three use the same frame structure for their link-layer frames as well. All three standards have the ability to reduce their transmission rate in order to reach out over greater distances. And all three standards allow for both “infrastructure mode” and “ad hoc mode,” as we’ll also shortly discuss. However, as shown in Table 6.1, the three standards have some major differences at the physical layer.

The 802.11b wireless LAN has a data rate of 11 Mbps and operates in the unlicensed frequency band of 2.4–2.485 GHz, competing for frequency spectrum with 2.4 GHz phones and microwave ovens. 802.11a wireless LANs can run at

| Standard | Frequency Range (United States) | Data Rate     |
|----------|---------------------------------|---------------|
| 802.11b  | 2.4–2.485 GHz                   | up to 11 Mbps |
| 802.11a  | 5.1–5.8 GHz                     | up to 54 Mbps |
| 802.11g  | 2.4–2.485 GHz                   | up to 54 Mbps |

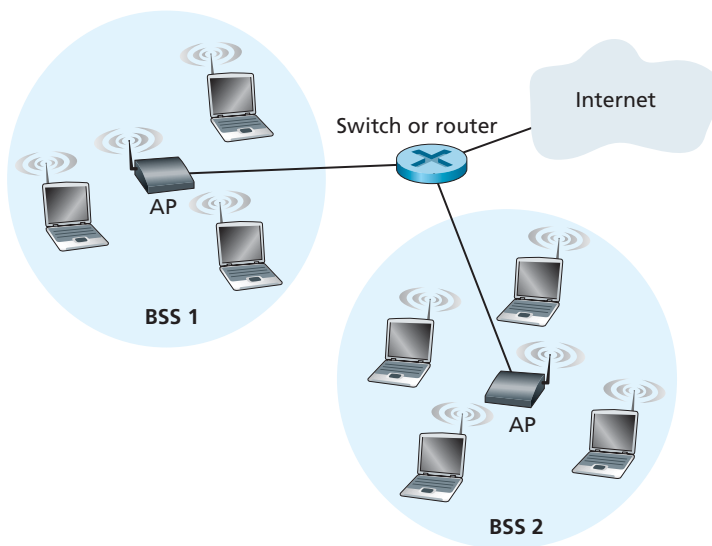
**Table 6.1** ♦ Summary of IEEE 802.11 standards

significantly higher bit rates, but do so at higher frequencies. By operating at a higher frequency, 802.11a LANs have a shorter transmission distance for a given power level and suffer more from multipath propagation. 802.11g LANs, operating in the same lower-frequency band as 802.11b and being backwards compatible with 802.11b (so one can upgrade 802.11b clients incrementally) yet with the higher-speed transmission rates of 802.11a, allows users to have their cake and eat it too.

A relatively new WiFi standard, 802.11n [IEEE 802.11n 2012], uses multiple-input multiple-output (MIMO) antennas; i.e., two or more antennas on the sending side and two or more antennas on the receiving side that are transmitting/receiving different signals [Diggavi 2004]. Depending on the modulation scheme used, transmission rates of several hundred megabits per second are possible with 802.11n.

### 6.3.1 The 802.11 Architecture

Figure 6.7 illustrates the principal components of the 802.11 wireless LAN architecture. The fundamental building block of the 802.11 architecture is the **basic service set (BSS)**. A BSS contains one or more wireless stations and a central



**Figure 6.7** ♦ IEEE 802.11 LAN architecture